

Assessment Test

Use the following assessment test to gauge your current level of skill in Java for the 1Z0-829. This test is designed to highlight some topics for your strengths and weaknesses so that you know which chapters you might want to read multiple times. Even if you do well on the assessment test, you should still read the book from cover to cover, as the real exams are quite challenging.

1. What is the result of executing the following code snippet?

```
41: final int score1 = 8, score2 = 3;
42: char myScore = 7;
43: var goal = switch (myScore) {
44:     default -> {if(10>score1) yield "unknown";}
45:     case score1 -> "great";
46:     case 2, 4, 6 -> "good";
47:     case score2, 0 -> {"bad";}
48: };
49: System.out.println(goal);
```

- A. unknown
- B. great
- C. good
- D. bad
- E. unknowngreatgoodbad
- F. Exactly one line needs to be changed for the code to compile.
- G. Exactly two lines need to be changed for the code to compile.**
- H. None of the above

2. What is the output of the following code snippet?

```
int moon = 9, star = 2 + 2 * 3;
float sun = star>10 ? 1 : 3;
double jupiter = (sun + moon) - 1.0f;
int mars = --moon <= 8 ? 2 : 3;
System.out.println(sun+", "+jupiter+", "+mars);
```

- A. 1, 11, 2
- B. 3.0, 11.0, 2**
- C. 1.0, 11.0, 3
- D. 3.0, 13.0, 3
- E. 3.0f, 12, 2
- F. The code does not compile because one of the assignments requires an explicit numeric cast.

3. Which changes, when made independently, guarantee the following code snippet prints 100 at runtime? (Choose all that apply.)

```
List<Integer> data = new ArrayList<>();
IntStream.range(0,100).parallel().forEach(s -> data.add(s));
System.out.println(data.size());
```

- A. Change data to an instance variable and mark it volatile.
- B. Remove `parallel()` in the stream operation.
- C. Change `forEach()` to `forEachOrdered()` in the stream operation.
- D. Change `parallel()` to `serial()` in the stream operation.
- E. Wrap the lambda body with a `synchronized` block.
- F. The code snippet will always print 100 as is.

4. What is the output of this code?

```
20: Predicate<String> empty = String::isEmpty;
21: Predicate<String> notEmpty = empty.negate();
22:
23: var result = Stream.generate(() -> "")
24:     .filter(notEmpty)
25:     .collect(Collectors.groupingBy(k -> k))
26:     .entrySet()
27:     .stream()
28:     .map(Entry::getValue)
29:     .flatMap(Collection::stream)
30:     .collect(Collectors.partitioningBy(notEmpty));
31: System.out.println(result);
```

- A. It outputs {}.
- B. It outputs {false=[], true=[]}.
- C. The code does not compile.
- D. The code does not terminate.

5. What is the result of the following program?

```
1: public class MathFunctions {
2:     public static void addToInt(int x, int amountToAdd) {
3:         x = x + amountToAdd;
4:     }
5:     public static void main(String[] args) {
6:         var a = 15;
7:         var b = 10;
8:         MathFunctions.addToInt(a, b);
9:         System.out.println(a);    } }
```

- A. 10
 - B. 15
 - C. 25
 - D. Compiler error on line 3
 - E. Compiler error on line 8
 - F. None of the above
6. Suppose that we have the following property files and code. What values are printed on lines 8 and 9, respectively?

Penguin.properties

name=Billy
age=1

Penguin_de.properties

name=Chilly
age=4

Penguin_en.properties

name=Willy

```
5: Locale fr = new Locale("fr");  
6: Locale.setDefault(new Locale("en", "US"));  
7: var b = ResourceBundle.getBundle("Penguin", fr);  
8: System.out.println(b.getString("name"));  
9: System.out.println(b.getString("age"));
```

- A. Billy and 1
 - B. Billy and null
 - C. Willy and 1
 - D. Willy and null
 - E. Chilly and null
 - F. The code does not compile.
7. What is guaranteed to be printed by the following code? (Choose all that apply.)

```
int[] array = {6,9,8};  
System.out.println("B" + Arrays.binarySearch(array,9));  
System.out.println("C" + Arrays.compare(array,  
    new int[] {6, 9, 8}));  
System.out.println("M" + Arrays.mismatch(array,  
    new int[] {6, 9, 8}));
```

- A. B1
 - B. B2
 - C. C-1
 - D. C0
 - E. M-1
 - F. M0
 - G. The code does not compile.
8. Which functional interfaces complete the following code, presuming variable `r` exists? (Choose all that apply.)
- ```
6: _____ x = r.negate();
7: _____ y = () -> System.out.println();
8: _____ z = (a, b) -> a - b;
```
- A. `BinaryPredicate<Integer, Integer>`
  - B. `Comparable<Integer>`
  - C. `Comparator<Integer>`
  - D. `Consumer<Integer>`
  - E. `Predicate<Integer>`
  - F. `Runnable`
  - G. `Runnable<Integer>`
9. Suppose you have a module named `com.vet`. Where could you place the following `module-info.java` file to create a valid module?
- ```
public module com.vet {
    exports com.vet;
}
```
- A. At the same level as the `com` folder
 - B. At the same level as the `vet` folder
 - C. Inside the `vet` folder
 - D. None of the above
10. What is the output of the following program? (Choose all that apply.)
- ```
1: interface HasTail { private int getTailLength(); }
2: abstract class Puma implements HasTail {
3: String getTailLength() { return "4"; }
4: }
5: public class Cougar implements HasTail {
6: public static void main(String[] args) {
7: var puma = new Puma() {};
```

```
8: System.out.println(puma.getTailLength());
9: }
10: public int getTailLength(int length) { return 2; }
11: }
```

A. 2

B. 4

C. The code will not compile because of line 1.

D. The code will not compile because of line 3.

E. The code will not compile because of line 5.

F. The code will not compile because of line 7.

G. The code will not compile because of line 10.

H. The output cannot be determined from the code provided.

11. Which lines in Tadpole.java give a compiler error? (Choose all that apply.)

```
// Frog.java
1: package animal;
2: public class Frog {
3: protected void ribbit() { }
4: void jump() { }
5: }

// Tadpole.java
1: package other;
2: import animal.*;
3: public class Tadpole extends Frog {
4: public static void main(String[] args) {
5: Tadpole t = new Tadpole();
6: t.ribbit();
7: t.jump();
8: Frog f = new Tadpole();
9: f.ribbit();
10: f.jump();
11: } }
```

A. Line 5

B. Line 6

C. Line 7

D. Line 8

E. Line 9

F. Line 10

G. All of the lines compile.

12. Which of the following can fill in the blanks in order to make this code compile?

```

_____ a = _____.getConnection(
 url, userName, password);
_____ b = a.prepareStatement(sql);
_____ c = b.executeQuery();
if (c.next()) System.out.println(c.getString(1));

```

- A. Connection, Driver, PreparedStatement, ResultSet
- B. Connection, DriverManager, PreparedStatement, ResultSet**
- C. Connection, DataSource, PreparedStatement, ResultSet
- D. Driver, Connection, PreparedStatement, ResultSet
- E. DriverManager, Connection, PreparedStatement, ResultSet
- F. DataSource, Connection, PreparedStatement, ResultSet

13. Which of the following statements can fill in the blank to make the code compile successfully? (Choose all that apply.)

```
Set<? extends RuntimeException> mySet = new _____ ();
```

- A. HashSet<? extends RuntimeException>
- B. HashSet<Exception>
- C. TreeSet<RuntimeException>**
- D. TreeSet<NullPointerException>**
- E. None of the above

14. Assume that `birds.dat` exists, is accessible, and contains data for a `Bird` object. What is the result of executing the following code? (Choose all that apply.)

```

1: import java.io.*;
2: public class Bird {
3: private String name;
4: private transient Integer age;
5:
6: // Getters/setters omitted
7:
8: public static void main(String[] args) {
9: try(var is = new ObjectInputStream(
10: new BufferedInputStream(
11: new FileInputStream("birds.dat")))) {
12: Bird b = is.readObject();
13: System.out.println(b.age);
14: } } }

```

- A. It compiles and prints 0 at runtime.
  - B. It compiles and prints null at runtime.
  - C. It compiles and prints a number at runtime.
  - D. The code will not compile because of lines 9–11.
  - E. The code will not compile because of line 12.
  - F. It compiles but throws an exception at runtime.
15. Which of the following are valid instance members of a class? (Choose all that apply.)
- A. `var var = 3;`
  - B. `Var case = new Var();`
  - C. `void var() {}`
  - D. `int Var() { var _ = 7; return _;}`
  - E. `String new = "var";`
  - F. `var var() { return null; }`
16. Which is true if the table is empty before this code is run? (Choose all that apply.)

```
var sql = "INSERT INTO people VALUES(?, ?, ?)";
conn.setAutoCommit(false);
```

```
try (var ps = conn.prepareStatement(sql,
 ResultSet.TYPE_SCROLL_SENSITIVE,
 ResultSet.CONCUR_UPDATABLE)) {
```

```
 ps.setInt(1, 1);
 ps.setString(2, "Joslyn");
 ps.setString(3, "NY");
 ps.executeUpdate();
```

```
 Savepoint sp = conn.setSavepoint();
```

```
 ps.setInt(1, 2);
 ps.setString(2, "Kara");
 ps.executeUpdate();
```

```
 conn._____;
}
```

- A. If the blank line contains `rollback()`, there are no rows in the table.
  - B. If the blank line contains `rollback()`, there is one row in the table.
  - C. If the blank line contains `rollback(sp)`, there are no rows in the table.
  - D. If the blank line contains `rollback(sp)`, there is one row in the table.
  - E. The code does not compile.
  - F. The code throws an exception because the second update does not set all the parameters.
17. Which is true if the contents of `path1` start with the text `Howdy`? (Choose two.)
- ```
System.out.println(Files.mismatch(path1,path2));
```
- A. If `path2` doesn't exist, the code prints `-1`.
 - B. If `path2` doesn't exist, the code prints `0`.
 - C. If `path2` doesn't exist, the code throws an exception.
 - D. If the contents of `path2` start with `Hello`, the code prints `-1`.
 - E. If the contents of `path2` start with `Hello`, the code prints `0`.
 - F. If the contents of `path2` start with `Hello`, the code prints `1`.
18. Which of the following types can be inserted into the blank to allow the program to compile successfully? (Choose all that apply.)
- ```
1: import java.util.*;
2: final class Amphibian {}
3: abstract class Tadpole extends Amphibian {}
4: public class FindAllTadpoles {
5: public static void main(String... args) {
6: var tadpoles = new ArrayList<Tadpole>();
7: for (var amphibian : tadpoles) {
8: _____ tadpole = amphibian;
9: } } }
```
- A. `List<Tadpole>`
  - B. `Boolean`
  - C. `Amphibian`
  - D. `Tadpole`
  - E. `Object`
  - F. None of the above
19. What is the result of compiling and executing the following program?
- ```
1: public class FeedingSchedule {
2:     public static void main(String[] args) {
3:         var x = 5;
4:         var j = 0;
```



```
5:         OUTER: for (var i = 0; i < 3;)
6:             INNER: do {
7:                 i++;
8:                 x++;
9:                 if (x > 10) break INNER;
10:                x += 4;
11:                j++;
12:            } while (j <= 2);
13:        System.out.println(x);
14:    } }
```

- A. 10
- B. 11
- C. 12**
- D. 17
- E. The code will not compile because of line 5.
- F. The code will not compile because of line 6.

20. When printed, which `String` gives the same value as this text block?

```
var pooh = ""
    "Oh, bother." -Pooh
    """.indent(1);
System.out.print(pooh);
```

- A. `"\n\"Oh, bother.\" -Pooh\n"`
- B. `"\n \"Oh, bother.\" -Pooh\n"`
- C. `" \"Oh, bother.\" -Pooh\n"`**
- D. `"\n\"Oh, bother.\" -Pooh"`
- E. `"\n \"Oh, bother.\" -Pooh"`
- F. `" \"Oh, bother.\" -Pooh"`
- G. None of the above

21. A(n) _____ module always contains a `module-info.java` file, while a(n) _____ module always exports all its packages to other modules.

- A. automatic, named
- B. automatic, unnamed
- C. named, automatic**
- D. named, unnamed
- E. unnamed, automatic
- F. unnamed, named
- G. None of the above

22. What is the result of the following code?

```
22: var treeMap = new TreeMap<Character, Integer>();
23: treeMap.put('k', 1);
24: treeMap.put('k', 2);
25: treeMap.put('m', 3);
26: treeMap.put('M', 4);
27: treeMap.replaceAll((k, v) -> v + v);
28: treeMap.keySet()
29:   .forEach(k -> System.out.print(treeMap.get(k)));
```

- A. 268
- B. 468
- C. 2468
- D. 826
- E. 846
- F. 8246
- G. None of the above

23. Which of the following lines can fill in the blank to print true? (Choose all that apply.)

```
10: public static void main(String[] args) {
11:     System.out.println(test(_____));
12: }
13: private static boolean test(Function<Integer, Boolean> b) {
14:     return b.apply(5);
15: }
```

- A. `i::equals(5)`
- B. `i -> {i == 5;}`
- C. `(i) -> i == 5`
- D. `(int i) -> i == 5`
- E. `(int i) -> {return i == 5;}`
- F. `(i) -> {return i == 5;}`

24. How many times is the word true printed?

```
var s1 = "Java";
var s2 = "Java";
var s3 = s1.indent(1).strip();
var s4 = s3.intern();
var sb1 = new StringBuilder();
sb1.append("Ja").append("va");
```

```
System.out.println(s1 == s2);
System.out.println(s1.equals(s2));
System.out.println(s1 == s3);
System.out.println(s1 == s4);
System.out.println(sb1.toString() == s1);
System.out.println(sb1.toString().equals(s1));
```

- A. Once
- B. Twice
- C. Three times
- D. Four times**
- E. Five times
- F. The code does not compile.

25. What is the output of the following program?

```
1: class Deer {
2:     public Deer() {System.out.print("Deer");}
3:     public Deer(int age) {System.out.print("DeerAge");}
4:     protected boolean hasHorns() { return false; }
5: }
6: public class Reindeer extends Deer {
7:     public Reindeer(int age) {System.out.print("Reindeer");}
8:     public boolean hasHorns() { return true; }
9:     public static void main(String[] args) {
10:         Deer deer = new Reindeer(5);
11:         System.out.println(", " + deer.hasHorns());
12: } }
```

- A. ReindeerDeer,false
- B. DeerAgeReindeer,true
- C. DeerReindeer,true**
- D. DeerReindeer,false
- E. ReindeerDeer,true
- F. DeerAgeReindeer,false
- G. The code will not compile because of line 4.
- H. The code will not compile because of line 12.

26. Which of the following are true? (Choose all that apply.)

```
private static void magic(Stream<Integer> s) {
    Optional o = s
        .filter(x -> x < 5)
        .limit(3)
        .max((x, y) -> x-y);
    System.out.println(o.get());
}
```

- A. `magic(Stream.empty());` runs infinitely.
 - B. `magic(Stream.empty());` throws an exception.
 - C. `magic(Stream.iterate(1, x -> x++));` runs infinitely.
 - D. `magic(Stream.iterate(1, x -> x++));` throws an exception.
 - E. `magic(Stream.of(5, 10));` runs infinitely.
 - F. `magic(Stream.of(5, 10));` throws an exception.
 - G. The method does not compile.
27. Assuming the following declarations are top-level types declared in the same file, which successfully compile? (Choose all that apply.)

```
record Music() {
    final int score = 10;
}
record Song(String lyrics) {
    Song {
        this.lyrics = lyrics + "Hello World";
    }
}
sealed class Dance {}
record March() {
    @Override String toString() { return null; }
}
class Ballet extends Dance {}
```

- A. Music
- B. Song
- C. Dance
- D. March
- E. Ballet
- F. None of them compile.

28. Which of the following expressions compile without error? (Choose all that apply.)

- A. `int monday = 3 + 2.0;`
- B. `double tuesday = 5_6L;`
- C. `boolean wednesday = 1 > 2 ? !true;`
- D. `short thursday = (short)Integer.MAX_VALUE;`
- E. `long friday = 8.0L;`
- F. `var saturday = 2_.0;`
- G. None of the above

29. What is the result of executing the following application?

```
final var cb = new CyclicBarrier(3,  
    () -> System.out.println("Clean!")); // u1  
ExecutorService service = Executors.newSingleThreadExecutor();  
try {  
    IntStream.generate(() -> 1)  
        .limit(12)  
        .parallel()  
        .forEach(i -> service.submit(() -> cb.await())); // u2  
} finally { service.shutdown(); }
```

- A. It outputs Clean! at least once.
- B. It outputs Clean! exactly four times.
- C. The code will not compile because of line u1.
- D. The code will not compile because of line u2.
- E. It compiles but throws an exception at runtime.
- F. It compiles but waits forever at runtime.

30. Which statement about the following method is true?

```
5: public static void main(String... unused) {  
6:     System.out.print("a");  
7:     try (StringBuilder reader = new StringBuilder()) {  
8:         System.out.print("b");  
9:         throw new IllegalArgumentException();  
10:    } catch (Exception e || RuntimeException e) {  
11:        System.out.print("c");  
12:        throw new FileNotFoundException();  
13:    } finally {  
14:        System.out.print("d");  
15:    } }
```

- A. It compiles and prints abc.
- B. It compiles and prints abd.
- C. It compiles and prints abcd.
- D. One line contains a compiler error.
- E. Two lines contain a compiler error.
- F. Three lines contain a compiler error.
- G. It compiles but prints an exception at runtime.

Respuestas del cuestionario semana 1	
Pregunta	Comentario
1	Creo que no compila porque algunos case del switch no devuelven un valor. Cuando un switch se usa para asignar algo, todos los caminos deberían devolver un resultado.
2	Primero se calcula $2 * 3$, entonces star vale 8. Como no es mayor que 10, sun vale 3. Después se hacen los demás cálculos y el resultado es 3.0, 11.0, 2.
3	El problema es que varios hilos están agregando datos al mismo ArrayList. Como no es seguro para concurrencia, el tamaño final puede no ser 100.
4	El stream genera cadenas vacías para siempre. Como el filtro busca cadenas que no estén vacías, nunca encuentra una y el programa no termina.
5	Java pasa una copia del valor al método. Aunque dentro del método cambie x, la variable a sigue siendo 15.
6	Como no existe un archivo para francés, Java usa el idioma por defecto y busca los valores en los archivos disponibles.
7	Los dos arreglos son iguales. Por eso compare() devuelve 0 y mismatch() devuelve -1. La búsqueda binaria no es segura porque el arreglo no está ordenado.
8	negate() pertenece a Predicate. La segunda expresión parece un Runnable y la tercera un Comparator.
9	Creo que el error está en que los módulos no se declaran usando public module.
10	Un método privado dentro de una interfaz debe tener cuerpo. Aquí sólo está declarado.
11	Hay problemas de acceso. Algunos métodos sólo pueden usarse desde el mismo paquete o desde una subclase.
12	Para hacer una consulta JDBC normalmente se obtiene una Connection, luego un PreparedStatement y después un ResultSet.
13	El wildcard indica que debe ser RuntimeException o alguna clase hija de ella.
14	Veo errores relacionados con excepciones y también falta convertir el resultado de readObject().
15	var puede usarse como nombre de método, pero no como tipo de atributo o tipo de retorno.
16	Si hago rollback completo se deshacen todos los cambios. Si vuelvo al savepoint sólo se deshacen los cambios posteriores.
17	Si el archivo no existe se produce una excepción. Si existen diferencias, devuelve la posición donde aparece la primera.
18	No compila porque una clase marcada como final no puede tener subclases.
19	Siguiendo los bucles paso a paso, el valor final de x termina siendo 12.
20	El text block tiene un salto de línea al final, pero no al principio.
21	Los módulos nombrados usan module-info.java y los módulos automáticos exportan todos sus paquetes.
22	Cuando se repite una clave en un Map, el valor anterior se reemplaza. Luego los valores se duplican y se imprimen ordenados.
23	Las expresiones lambda válidas son las que reciben un valor y devuelven un boolean.
24	Algunas comparaciones revisan referencias y otras revisan contenido. Por eso aparecen cuatro resultados true.
25	Primero se ejecuta el constructor de la clase padre y luego el de la hija. Después se usa el método sobrescrito.
26	Si el Optional está vacío y llamamos a get(), se lanza una excepción.

27	Hay varios errores con los records y con las clases selladas. La única opción válida es la clase Dance.
28	Algunas expresiones fallan por tipos incompatibles y otras funcionan porque Java puede hacer conversiones automáticamente.
29	El CyclicBarrier espera tres hilos, pero sólo hay uno disponible. Por eso el programa se queda esperando.
30	Veo varios errores de compilación. El StringBuilder no sirve para try-with-resources, el catch está mal escrito y además hay una excepción sin manejar.